

TrES-3b

R-2722

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_220730 · created 2026-09-01T23:15:33+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459790.7587 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.192 +/- 0.048
Transit depth (Rp/Rs)^2	3.67 +/- 1.85 [%]
Orbital Inclination (inc)	81.29 +/- 2.31
Airmass coefficient 1 (a1)	1.02 +/- 0.025
Airmass coefficient 2 (a2)	-0.017 +/- 0.038
Scatter in the residuals of the lightcurve fit is	2.46 %
Best Comparison Star	#3 - [276, 198]
Optimal Method	PSF photometry
Transit Duration (day)	0.045 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.192 ± 0.048 vs 0.16309 (deviation 0.6σ)
Duration fit vs published	0.045 d vs 0.0567 d (deviation 0.43σ)
Mid-time O–C	2.4 min
Depth SNR	4.0
Residual scatter	2.46 %

Light curve

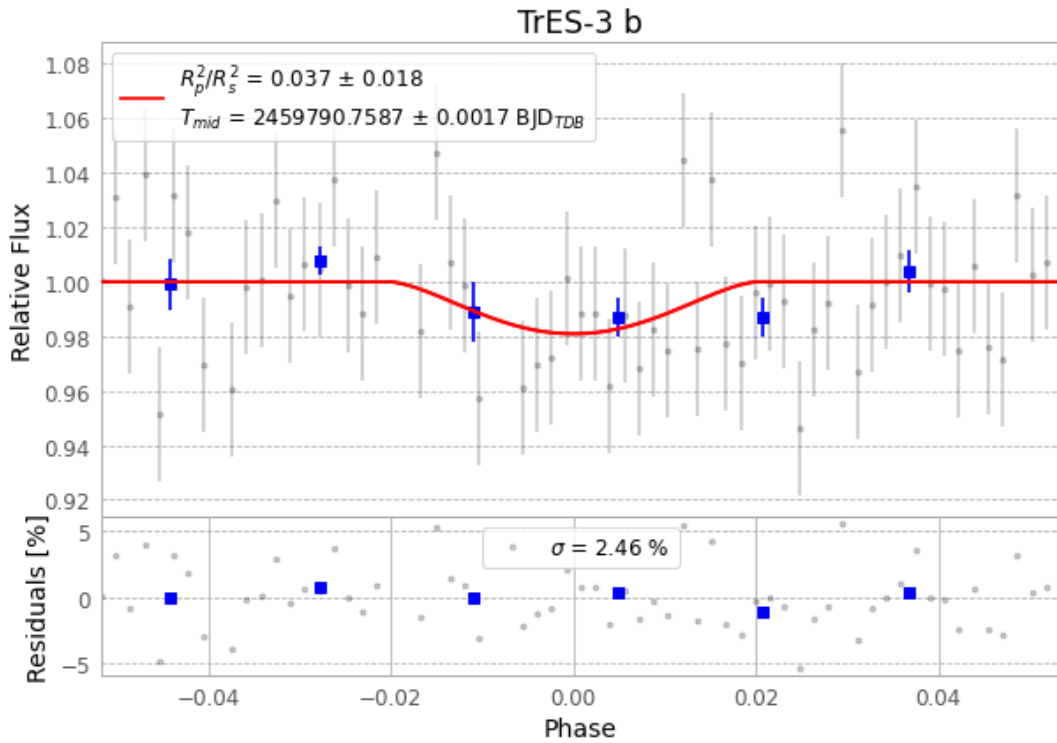


Figure 1 — FinalLightCurve_TrES-3 b_2022-07-29.png (SHA-256 7a05f0607a1d155f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [346, 359], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 cf36b7f9c88f6884...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

65 FITS frames (43 MB), TRES-3220730043013.FITS → TRES-3220730074813.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-220730.log (04b6896bc959...), FinalLightCurve_TrES-3 b_2022-07-29.png (7a05f0607a1d...), FinalLightCurve_TrES-3 b_2022-07-29.csv (2d1dea774e82...)

Compute resources

Wall time 151.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-01 23:15Z.